

HOMIBEE

Mission Briefing & System Initialization

An intelligent, event-driven architecture for autonomous home orchestration



Architecture

Agents

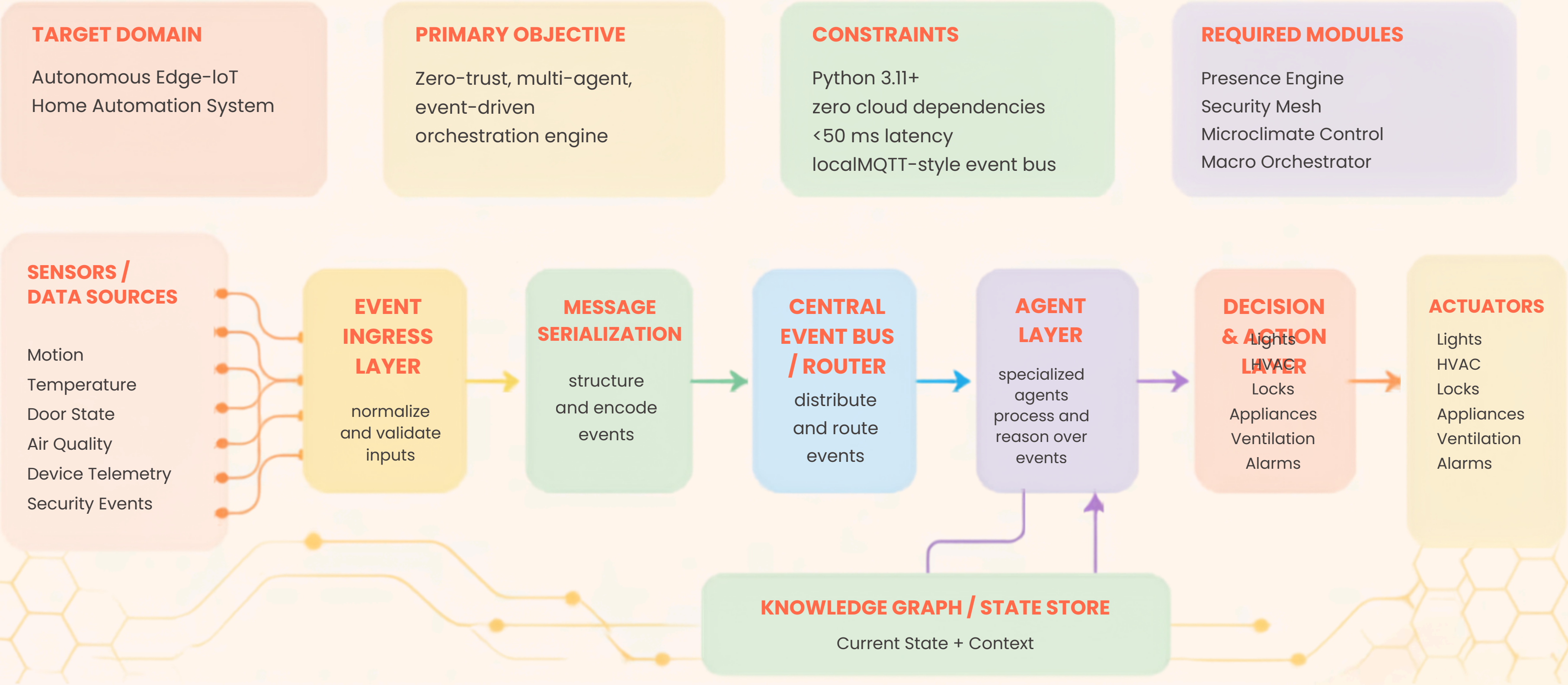
State Management

Real-Time Automation

Mission: From Sensors to Intelligent Action



System Architecture: The HOMIBEE Nervous System



Agent Ecosystem: Specialized Intelligence

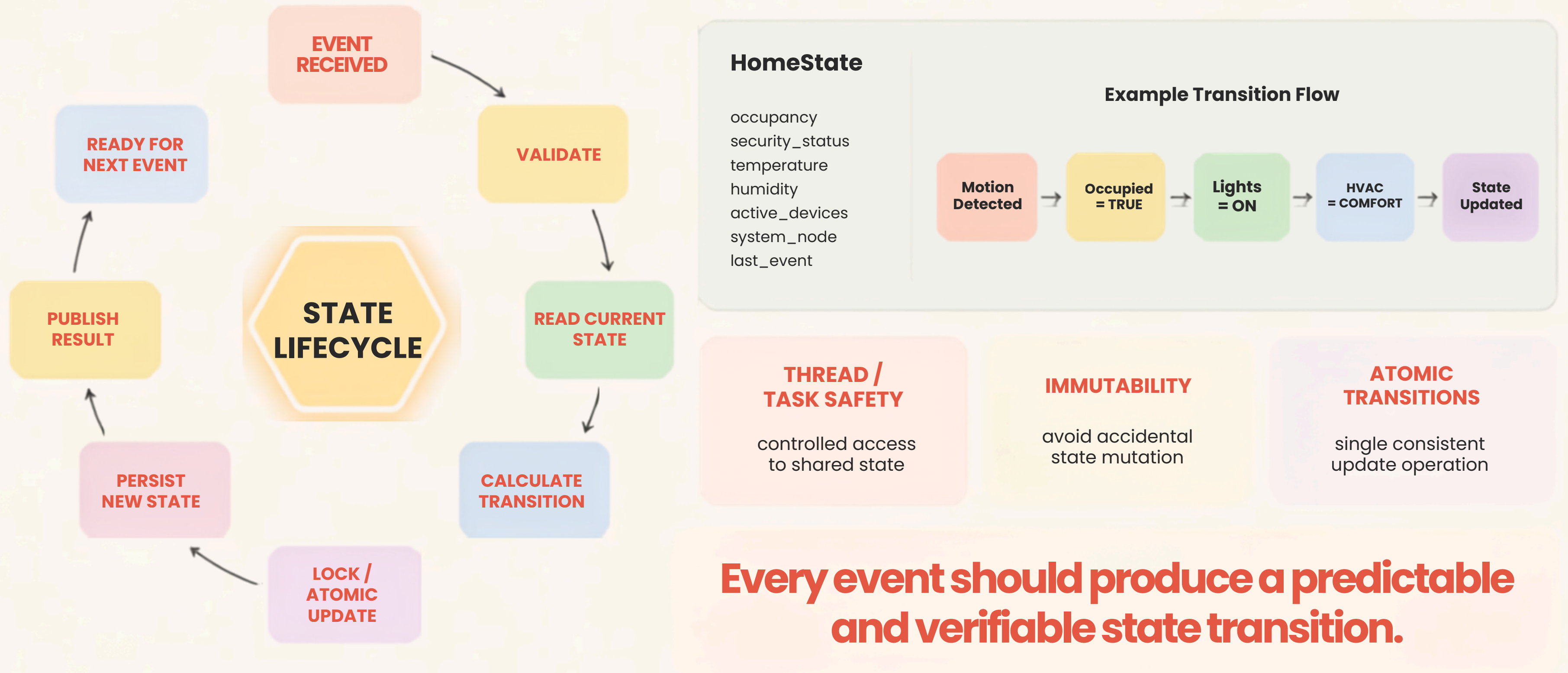
Agent	Primary Objective	Inputs / Tiggers	Published Actions
Presence Engine	Detect occupancy and activity	Motion, door sensors, room activity	Lighting, HVAC mode, occupancy state
Security Mesh	Detect/respond to security events	Door, window, motion, access events	Alarm, lock, security alerts
Microclimate Control	Maintain environmental comfort	Temperature, humidity, air quality	HVAC, fan, purifier commands
Macro Orestrator	Coordinate system-wide decisions	Agent events + global state	Priority decisions, workflows, policies



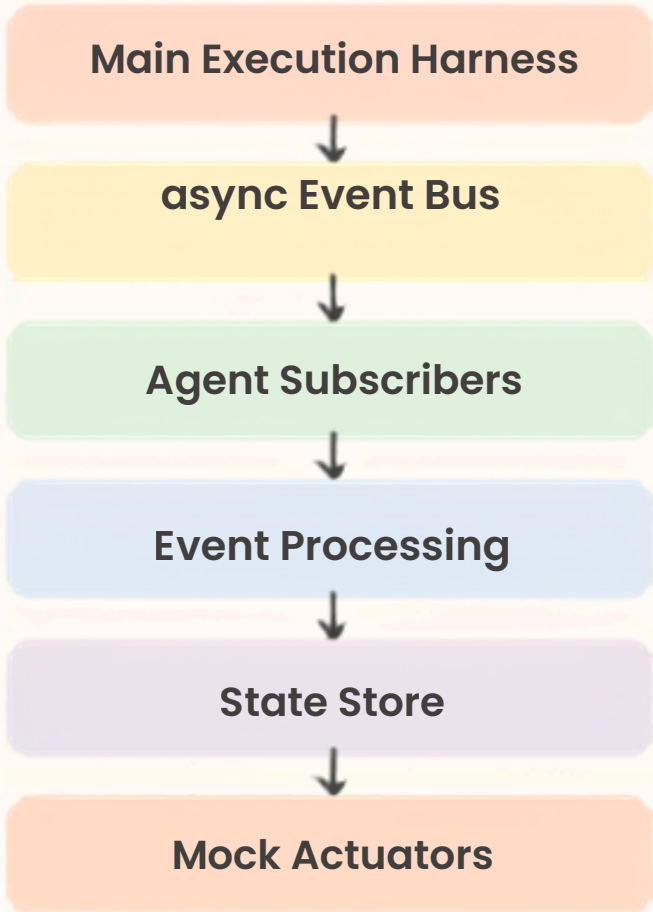
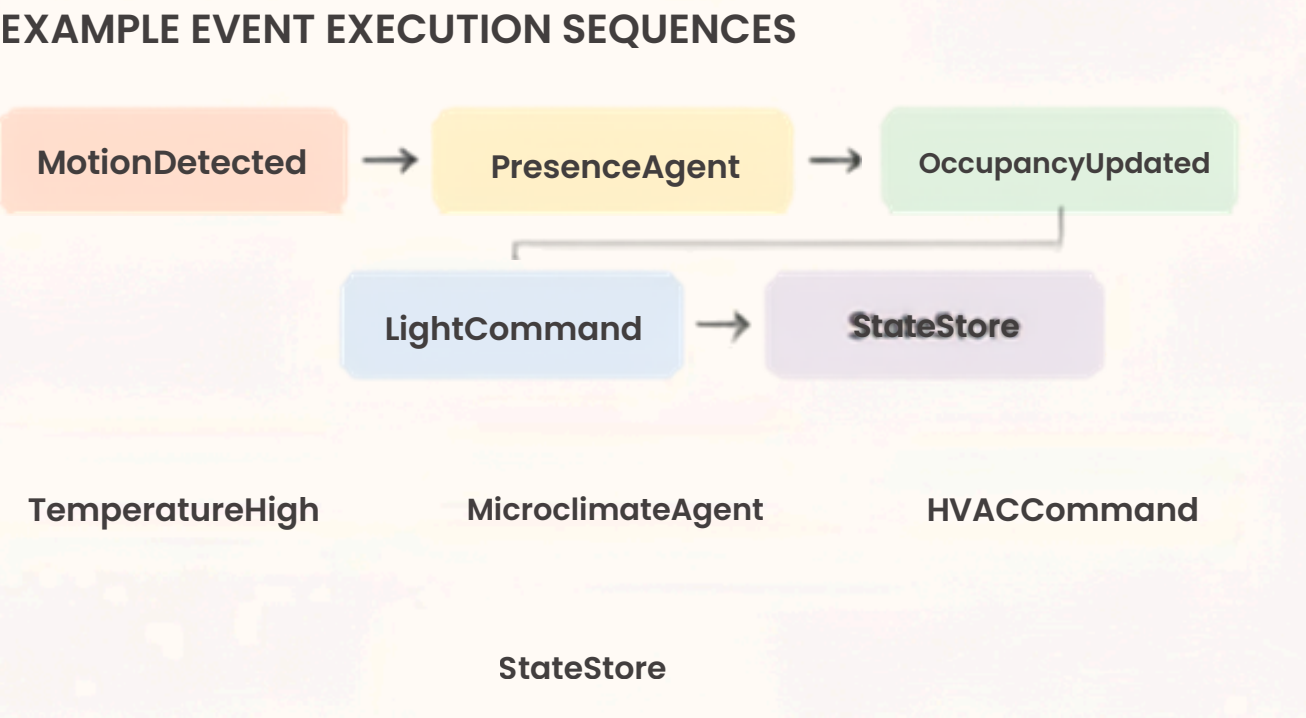
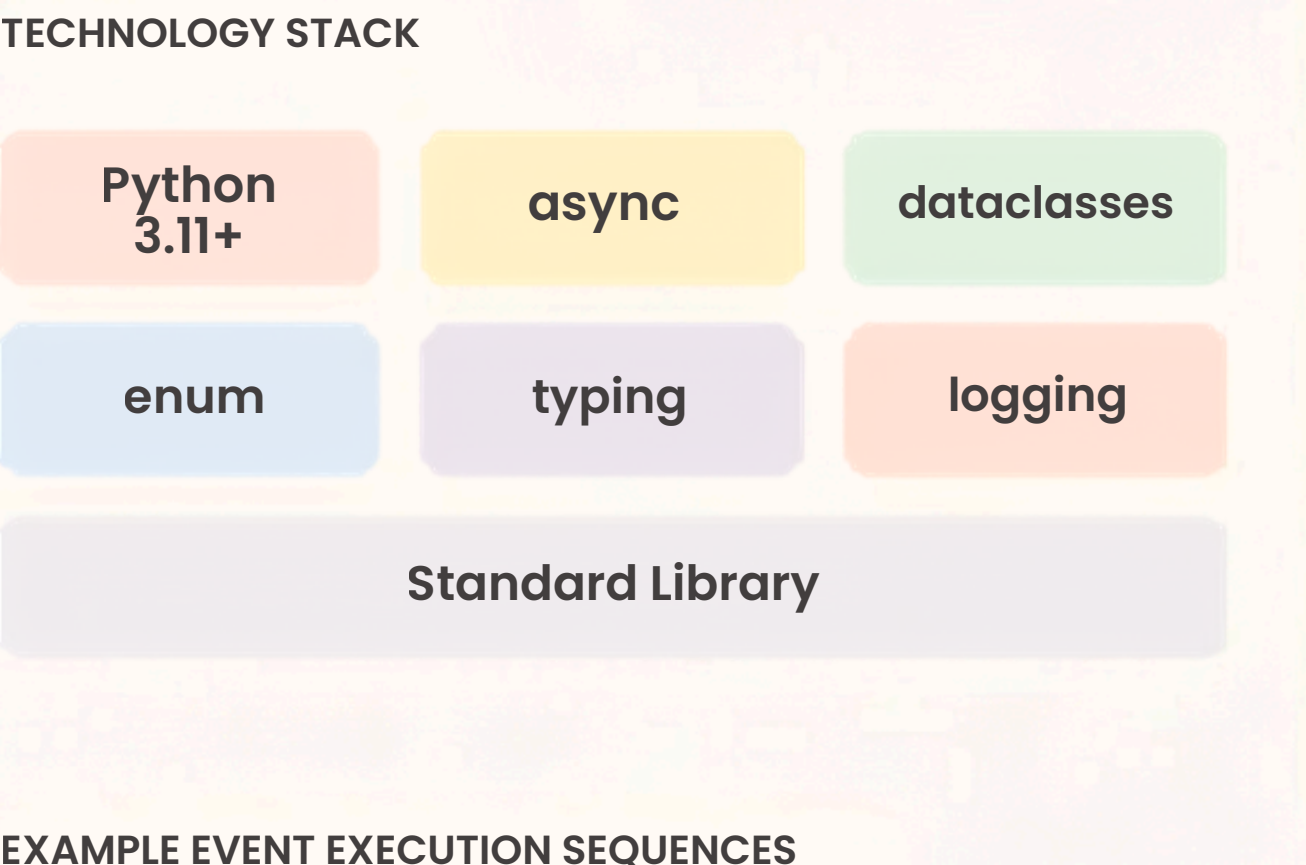


Agents are **isolated** in responsibility but **connected** through a shared event-driven communication layer.

Dynamic State: Keeping the System Consistent



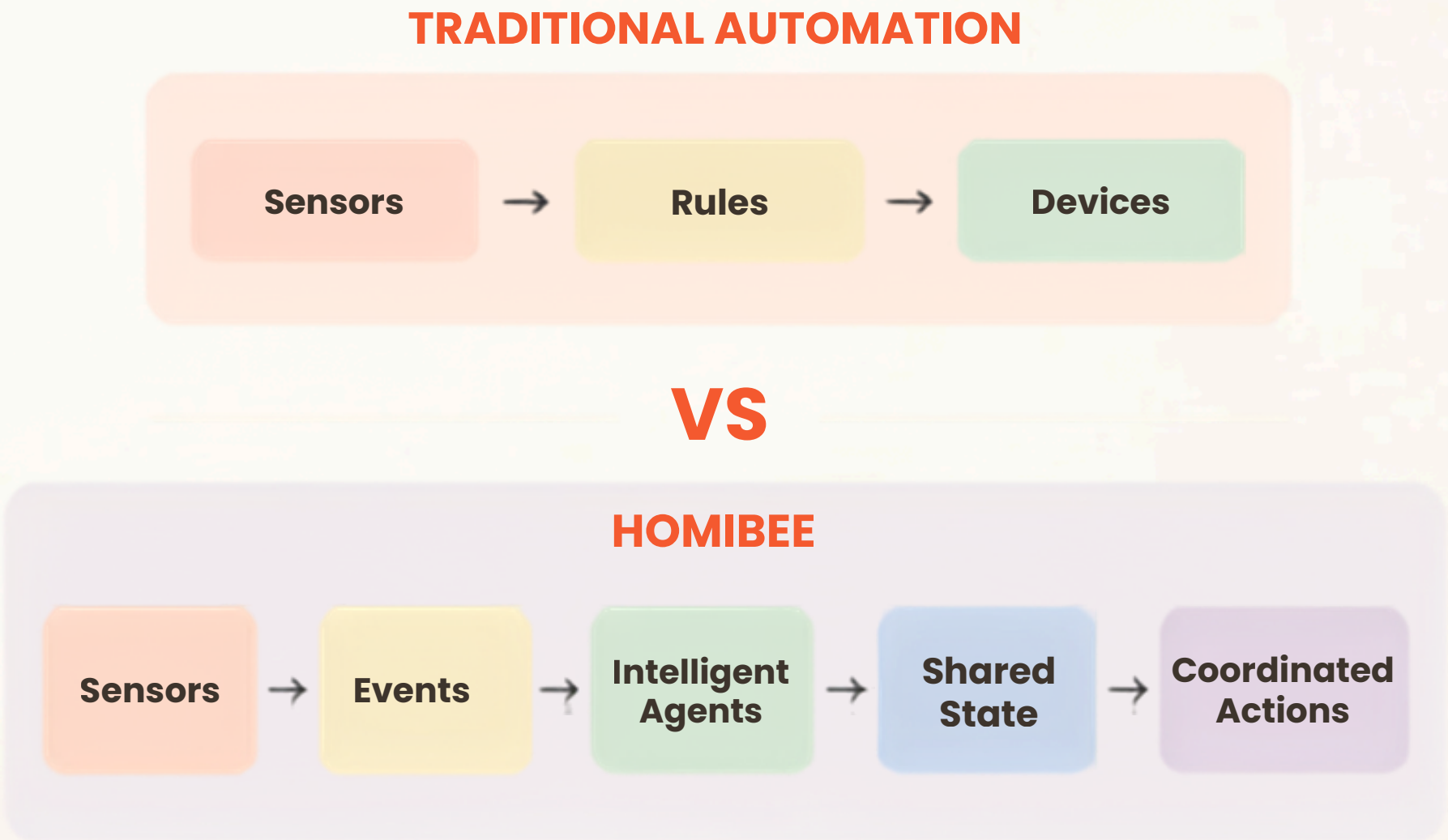
Implementation: From Architecture to Running Prototype



EVENT PROCESSING (PYTHON)

```
event = await bus.receive()
agent = registry.resolve(event)
new_state = await agent.process(event, state_store(snapshot))
await state_store.commit(new_state)
await bus.publish(result)
```

HOMIBEE: Expected Benefits & System Impact



HOMIBEE transforms disconnected smart-home events into coordinated, verifiable intelligence.

SENSE



COORDINATE



DECIDE



ACT